

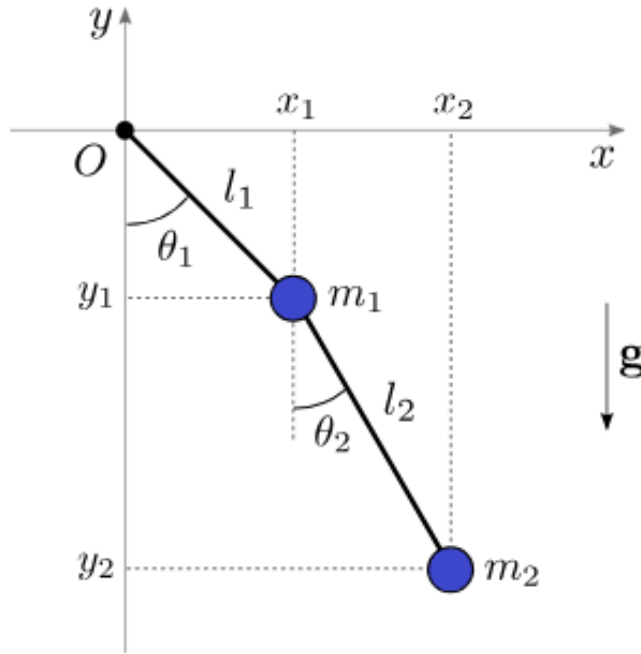


Figura 1: Diagram (from Diego Assencio)

For the following deduction we will assume the simplification thus  $l_1 = l_2 \equiv l$  and  $m_1 = m_2 \equiv m$ . Also, we will be using the following angles to simplify the reading between this paper's results and the Python code,  $\alpha = \theta_1$  and  $\beta = \theta_2$ .

First of all, we must our cartesian coordinates in terms of our two generalized ones.

$$\begin{aligned} x_1 &= l \sin \alpha & x_2 &= l (\sin \alpha + \sin \beta) \\ y_1 &= -l \cos \alpha & y_2 &= -l (\cos \alpha + \cos \beta) \end{aligned}$$

We can also get the velocities, which are just the coordinates time derivatives.

$$\begin{aligned} \dot{x}_1 &= \dot{\alpha} l \cos \alpha & \dot{x}_2 &= l (\dot{\alpha} \cos \alpha + \dot{\beta} \cos \beta) \\ \dot{y}_1 &= \dot{\alpha} l \sin \alpha & \dot{y}_2 &= l (\dot{\alpha} \sin \alpha + \dot{\beta} \sin \beta) \end{aligned}$$

Their square value will also come in handy to easily replace them in the kinetic energy formula.

$$\begin{aligned} \dot{x}_1 &= \dot{\alpha} l \cos \alpha & \dot{x}_2 &= l (\dot{\alpha} \cos \alpha + \dot{\beta} \cos \beta + \dot{\alpha} \dot{\beta} \cos \alpha \cos \beta) \\ \dot{y}_1 &= \dot{\alpha} l \sin \alpha & \dot{y}_2 &= l (\dot{\alpha} \sin \alpha + \dot{\beta} \sin \beta + \dot{\alpha} \dot{\beta} \sin \alpha \sin \beta) \end{aligned}$$

Using the well know formulas for the kinetic and gravitational potential energies we can define the Lagrangian of our system.

$$K = \frac{1}{2} m (\dot{x}_1^2 + \dot{y}_1^2 + \dot{x}_2^2 + \dot{y}_2^2) = ml\dot{\alpha}^2 + \frac{1}{2} ml\dot{\beta}^2 + ml\dot{\alpha}\dot{\beta} \cos(\alpha - \beta)$$

$$U = mg(y_1 + y_2) = -2mgl \cos \alpha - mgl \cos \beta$$

$$\Rightarrow L \equiv K - U = ml\dot{\alpha}^2 + \frac{1}{2} ml\dot{\beta}^2 + ml\dot{\alpha}\dot{\beta} \cos(\alpha - \beta) + 2mgl \cos \alpha + mgl \cos \beta$$

The equations of movement are given by Lagrange's Equation's solutions. In this system both lagrange equations are coupled and are  $\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{\alpha}} \right) - \frac{\partial L}{\partial \alpha} = 0$  and  $\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{\beta}} \right) - \frac{\partial L}{\partial \beta} = 0$ . We can first substitute all the terms in the first equation and simplify it.

$$\begin{cases} \frac{\partial L}{\partial \alpha} = 2ml\dot{\alpha} + ml\dot{\beta} \cos(\alpha - \beta) \\ \frac{d}{dt} \left( \frac{\partial L}{\partial \dot{\alpha}} \right) = 2ml\ddot{\alpha} + ml\ddot{\beta} \cos(\alpha - \beta) - ml\dot{\beta} \sin(\alpha - \beta)(\dot{\alpha} - \dot{\beta}) \\ \frac{\partial L}{\partial \alpha} = -ml\dot{\alpha}\dot{\beta} \sin(\alpha - \beta) - 2mgl \sin \alpha \end{cases} \Rightarrow \boxed{2\ddot{\alpha} + \ddot{\beta} \cos(\alpha - \beta) + \dot{\beta} \sin(\alpha - \beta) + 2\frac{g}{l} \sin \alpha = 0}$$

By doing the same with the second equation,

$$\begin{cases} \frac{\partial L}{\partial \dot{\alpha}} = ml\dot{\beta} + ml\dot{\alpha}\cos(\alpha - \beta) \\ \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\alpha}}\right) = ml\ddot{\beta} + ml\ddot{\alpha}\cos(\alpha - \beta) - ml\dot{\alpha}\sin(\alpha - \beta)(\dot{\alpha} - \dot{\beta}) \\ \frac{\partial L}{\partial \alpha} = -ml\dot{\alpha}\dot{\beta}\sin(\alpha - \beta) - mgl\sin\beta \end{cases} \Rightarrow \boxed{\ddot{\beta} + \ddot{\alpha}\cos(\alpha - \beta) - \dot{\alpha}\sin(\alpha - \beta) + \frac{g}{l}\sin\beta = 0}$$

To apply Euler's numeric method we need to find  $\frac{d\dot{\alpha}}{dt}$  and  $\frac{d\dot{\beta}}{dt}$ , which are just  $\ddot{\alpha}$  and  $\ddot{\beta}$  respectively. We can get  $\ddot{\alpha}$  with just angles and angular velocities by substituting  $\ddot{\beta}$  from the second equation into the first one.

$$\begin{aligned} \ddot{\beta} &= -\ddot{\alpha}\cos(\alpha - \beta) + \dot{\alpha}\sin(\alpha - \beta) - \frac{g}{l}\sin\beta \\ \Rightarrow 2\ddot{\alpha} + \left(-\ddot{\alpha}\cos(\alpha - \beta) + \dot{\alpha}\sin(\alpha - \beta) - \frac{g}{l}\sin\beta\right)\cos(\alpha - \beta) + \dot{\beta}\sin(\alpha - \beta) + 2\frac{g}{l}\sin\alpha &= 0 \\ \rightarrow \ddot{\alpha}(2 - \cos(\alpha - \beta)) + \sin(\alpha - \beta)(\dot{\alpha}\cos(\alpha - \beta) + \dot{\beta}) + \frac{g}{l}(2\sin\alpha - \cos(\alpha - \beta)\sin\beta) & \end{aligned}$$

By isolating  $\ddot{\alpha}$  and using the numeric result to substitute it into  $\ddot{\beta}$ , we already have both function we need to apply Euler's method.

$$\boxed{\frac{d\dot{\alpha}}{dt} = \left[\sin(\alpha - \beta)(\dot{\alpha}\cos(\alpha - \beta) + \dot{\beta}) + \frac{g}{l}(2\sin\alpha - \cos(\alpha - \beta)\sin\beta)\right] \frac{1}{\cos(\alpha - \beta) - 2}}$$

$$\boxed{\frac{d\dot{\beta}}{dt} = -\ddot{\alpha}\cos(\alpha - \beta) + \dot{\alpha}\sin(\alpha - \beta) - \frac{g}{l}\sin\beta}$$